Isaac Tunney

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Postdoctoral Researcher in Aerial Robotics
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EDUCATION & POSTDOCTORAL TRAINING

Postdoctoral in Mechanical Engineering 2026-present
Université de Sherbrooke

PhD in Mechanical Engineering (Fast-tracked from master's program) – GPA: 4.3/4.3 2021-2026
Université de Sherbrooke

Bachelor's degree in Mechanical Engineering (Co-op program) – GPA: 4.01/4.3 2016-2020
Concentration in aeronautics
Université de Sherbrooke

POSTDOCTORAL PROJECT 1 (2026)

Objective: *Develop a semi-active* ***self-levelling landing gear*** *for a 30-kg tree-planting UAV operating on uneven, inclined forestry terrain within a 6-month timeframe.*

Key Contributions:

- **Authored the winning proposal** for SPADE Open Call #2 (Horizon Europe), ranking **first among 72 teams** internationally and securing **CAD 100,000 in funding.**
- **Led a 4-person multidisciplinary team** responsible for PCB design, ROS integration, embedded software, mechanical design, manufacturing, and client communications with SPADE/NIBIO in Norway.
- **Designed all main components of the novel landing gear**, including rack-and-pinion transmission, servo-actuated friction brakes, and structural components, and integrated encoder feedback.
- **Developed multi-level control algorithms** to adjust leg friction for height and attitude control of the four-legged UAV.
- **Validated self-levelling** on uneven terrain with slopes up to 20%, achieving settling times below 3 s, attitude errors below 1°, and height errors below 0.5 cm.
- **Field-tested** the system during **tree-planting operations** in Norway.

DOCTORAL PROJECT (2021-2026)

Objective: *Develop a landing technology combining* ***friction-based shock absorbers****,* ***reverse thrust*** *and specialized* ***spiny feet*** *to enable small drones to land rapidly and reliably in* ***extreme outdoor conditions****.*

Key Contributions:

- Designed a novel landing gear using friction-based shock absorbers for **high-speed impacts** (<4 m/s).
- Developed custom landing algorithms in **C++ (ArduPilot)**, integrating reverse thrust to prevent flipping.
- Sized and built a **high-power UAV** (thrust-to-weight ratio of 8) designed for speed and maneuverability.
- Developed a **Monte Carlo simulation framework** with a 2D UAV model to predict landing performance.
- Designed **specialized feet** with **retractable spines** to enable landings on steep, icy surfaces like icebergs.
- **Demonstrated landings** on rooftops inclined up to 60°, ground vehicles travelling at 110 km/h, boats at 30 km/h, and iceberg slopes up to 60° in Iceland.
- **Demonstration videos** (YouTube links): Rooftops · High-speed vehicles · Boats · Icebergs

ACROBATIC ROBOT GRIPPERS (2024–2025)

Objective: *Develop robotic grippers and their underlying principles enabling acrobatic robots to repeatedly transfer between trapeze bars while withstanding high impact loads.*

Key Contributions:

- **Led an 18-month research project** alongside my doctoral research, with CAD 100,000+ in funding from a major U.S. robotics company, recruiting and supervising a master's student.
- **Designed & built** a **robotic hand** for repeated bar-to-bar transfers in <1.2 s, with impact forces of >30 g.
- Performed **dynamic simulations** in WM2D to understand key challenges; iterated over 50 mechanisms.
- Delivered a **functional prototype** with single-solenoid release, rapid cable-driven grasping, spring-driven passive resetting in under 0.75 s, and mechanical advantage above 25 for releases under high loads.

PUBLICATIONS & TALKS

Research Publications and Thesis:

- **I. Tunney**, J. Bass and A. L. Desbiens, "The Ice Dart: Landing on Steep Ice Using Spiny Feet, Dissipative Landing Gear and Reverse Thrust," *IEEE Transactions on Field Robotics*, 2026.
- **I. Tunney**, J. Bass and A. L. Desbiens, "Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on High-Speed Vehicles," *Journal of Field Robotics*, 2025.
- J. Bass, **I. Tunney** and A. L. Desbiens, "Adaptative Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on Inclined Surfaces," *IEEE Robotics and Automation Letters*, 2022.
- **I. Tunney**, "Friction shock absorbers and reverse thrust for fast UAV landings in extreme outdoor conditions," PhD thesis, *Université de Sherbrooke*, 2026.
- R. Boivin, **I. Tunney** and A. L. Desbiens, "Design and Validation of a High-Load & High-Speed Robotic End-Effector for Dynamic Trapeze Maneuvers," submitted to *Mechatronics*, 2026.
- **I. Tunney**, J. Bass, S. Lloyd, T. K. Opstrup and A. L. Desbiens, "BASiL: A Lightweight Brake-Adjusted Self-Levelling Landing Gear for Uneven and Inclined Terrain," *manuscript in preparation*.

Invited talks & conference presentations:

- *ArduPilot 2026 Conference:* Invited 30-min talk (invitation-only), Ottawa, October 2026 (*forthcoming*)
- *Carleton University, NSERC CREATE UTILI Workshop:* Invited 15-min seminar, "Friction Shock Absorbers and Reverse Thrust for Fast UAV Landing on Icebergs and Glaciers," 2025
- *IROS 2022:* Conference presentation, "Adaptative Friction Shock Absorbers and Reverse Thrust for Fast Multirotor Landing on Inclined Surfaces," Kyoto, Japan, 2022

Media outreach:

- *Global News:* Interview and filmed demonstration of the Ice Dart drone's landing capabilities, June 2026
- *Radio-Canada:* Live radio interview discussing the Ice Dart research, June 2026

SELECTED ACADEMIC PROJECTS

Senior design project – AeroStrabe: Single-Seat Flight Simulator for Urban Air Mobility 2019-2020

- Developed a Python **flight-dynamics model** and a C++ **motion-control software.**
- Integrated *Oculus VR headset* and *Unreal Engine* within a **real-time software architecture** synchronizing all simulator subsystems.
- Secured **USD 10,000** from *Epic Games*' MegaGrants program for innovation in virtual reality
- Project completed by **six students** in **12 months** with a CAD 50,000 budget
- Demo video**:** https://www.facebook.com/watch/?v=3886459841378560

5th semester integration project: Motorized patient-lifting system for care centers 2018

- Designed a **gripper** for **patient-lifting blankets**; **2nd place** in the *Createk funding competition*.

WORK EXPERIENCE

Walt Disney Imagineer, Disney Research (California, US) – 5-month internship 2023

- Designed and built a **2-DOF acrobatic robot** for mid-air maneuvers using fast-prototyping tools.
- Developed **multithreaded Python software** on a Raspberry Pi CM4, coordinating real-time communication between the robot, pendulums, and motion-capture system.
- **Simulated robot dynamics and control** in **Simscape Multibody** (Matlab) before implementation.
- **Fused** IMU and motion-capture **data** for real-time tracking.

Software specialist in flight aerodynamics, *CAE* (Montreal, QC) – 4-month internship 2019

- Developed and implemented flight dynamics programs in C++ for the Boeing B747 Simulator
- Automated flight database updates with Python, reducing task time from >35 h to <1 min

Aerial Robotics Designer, *Outreach Robotics* / *Createk* (Sherbrooke, QC) – 4-month internship 2019

- Designed 4 core mechanical & robotic components, from concept to prototype: main grippers to grasp branches, orientation-locking serrated joint, emergency release, and full structural assembly
- Contributed to the *DeLeaves* product now commercialized by *Outreach Robotics*

Staff Coordinator, Department of Production, *ArcelorMittal Produits Longs Canada* (Contrecœur, QC) – 4-month internship 2018

- Managed day-to-day operations of the steel casting department, supervising ~30 employees

Mechanical Engineering Intern, *Cycles Devinci* (Chicoutimi, QC) – 4-month internship 2017

- Designed and built an automated corrosion bath to test anti-corrosion products on bike parts

AWARDS, HONOURS AND FUNDING

- PhD thesis nominated for Thesis of the Year award at the Université de Sherbrooke 2026
- Most-read paper in the last 5 years of the Journal of Field Robotics (Observed on 10/10/2026) 2026
- Recipient of the CAMAQ scholarship for the next generation in aeronautic 2024
- First place in the CIADI capstone project pitch competition at Concordia University; awarded cash prize and an invitation to the 2024 ICAO Symposium on Advanced Air Mobility 2024
- UTILI Scholarship / member of the UTILI program 2023
- Certificates of academic excellence (top 10%) from the Université de Sherbrooke 2019, 2020
- NSERC and FRQNT scholarships for research internships at the Createk Design Lab 2019

SKILLS & CERTIFICATIONS

Languages	French (first language), English (oral 4.5/5, written 5/5)
Software	• Programming: C++, Python, MATLAB, Simulink, Arduino / Windows and Linux (Ubuntu) • Design: AutoCAD, Onshape, SolidWorks • Dynamics/Robotics: Motion Genesis, Gazebo Simulator, ROS 2, Simscape, WM2D
Certifications	• Canadian pilot certificate for basic drone operations – 40h+ of flight time with multirotors • EASA pilot license for Iceland's A1/A3 drone category – 5h of flight time with multirotors

SELECTED ADDITIONAL PROJECTS AND INTERESTS

- Simulation of two drones (using ArduPilot) controlled simultaneously with ROS in the Gazebo Simulator
- RRT motion planning algorithm and minimum snap trajectory generation for a quadcopter in MATLAB
- Omnidirectional rolling robot that plays and interacts with dogs, launches treats and throws balls (ongoing)

Personal interests: rock climbing, running, tennis, coffee / latte art design, and everything related to robots!